Sentence Boundaries Are Part of the Watermark Interface: A SWAN Audit and an Exact Resegmentation Framework

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Abstract

Resegmentation is a recognized weakness of sentence-level semantic watermarks. We audit this known risk and study exact evaluation at behavioral, algorithmic, and systems levels. First, we audit SWAN. On 30 saved five-unit texts, replacing four sentence-final punctuation marks with semicolons while preserving lexical tokens and order changed the number of below-threshold decisions from 1/30 to 30/30; restoring the saved five spans returned it to 1/30. This establishes a reversible short-text interface dependence, but a one-unit score ceiling implied by SWAN's existing statistic already explains the fixed-threshold outcome, and two normal-length count-neutral routes stopped before detection. Second, we formalize fixed-token, fixed-unit-count boundary relocation and instantiate exact dynamic programs for independent-span and first-order adjacent-span detectors using standard optimal-substructure arguments. The pairwise program agrees with brute force on 7,669 systematic and random instances, agrees with the additive program on 1,000 reductions, and solves a 300-token case containing 14,606,509 legal segmentations in 0.8054 seconds. A compiled factorized implementation exactly processes a synthetic real-data shape representing 3.94×10^{19} segmentations. Third, we stress-test the full path on SemStamp and SimMark. A final frozen SemStamp detector-health run obtained 4.7% test FPR and 95.4% positive TPR, but exact-certificate readiness stopped before scoring because one document required 78.6 million pruned span pairs and the two-replay budget was 86.65 GPU-hours. We therefore report no real certified rate. Together, the results show why segmentation must be specified as part of the detector contract, what exact certification can guarantee, and where an audited implementation still becomes impractical.

1 Introduction

Semantic text watermarks aim to preserve detectability under lexical or syntactic variation by placing signals in sentence-level representations. This design introduces an interface that token-level evaluations can obscure: before a sentence-level detector can score a document, some procedure must decide what its units are. The same word sequence may consequently induce different statistical trials, different encoder or parser calls, and different pairwise relations.

This risk has already been recognized. SWAN explicitly identifies splitting and merging sentences to alter AMR structure as a future challenge and suggests multi-sentence AMRs and subgraph signals (Ye et al., 2026). AliMark analyzes structural perturbations and uses reconstruction and bit-sequence alignment to recover detection (Li et al., 2026); SwordStamp studies resegmentation, unit-count effects, and sub-sentence units (Diaa et al., 2026). Our question is therefore how to audit a specific instance of that known risk and state the coverage of an exact evaluation. We do not claim to discover segmentation sensitivity.

We investigate this problem through one research program with two evidence streams. The first is an empirical audit of SWAN, an AMR-based semantic watermark (Ye et al., 2026). The second asks whether robustness to a declared set of word-preserving resegmentations can be certified exactly for sentence-level detectors such as SemStamp (Hou et al., 2024a), k-SemStamp (Hou et al., 2024b), and SimMark (Dabiriaghdam & Wang, 2025). These streams address the same interface at different levels, but their evidence is not

interchangeable: a SemStamp certificate does not establish a SWAN mechanism, and a SWAN counterexample does not establish that a certificate is non-vacuous.

The SWAN audit begins with a reversible behavior observation. Thirty saved texts contain five detection units each. Replacing four sentence-final marks with semicolons preserves lexical tokens and their order while making the default detector see one unit. Below-threshold decisions change from 1/30 to 30/30. Supplying the detector with the five saved spans restores the count to 1/30. However, with SWAN's default null rate, a one-unit document cannot cross the strict public threshold even if its unit is marked. The result quantifies a reversible interface behavior in the saved configuration; it does not establish that AMR semantics are uniquely fragile.

We attempted to isolate effects beyond unit-count reduction through normal-length, count-neutral boundary exchange. Two preregistered generation routes stopped before transformation and detection. A later targeted construction produced one detectable 126-word document, but not the source-anchored, independently reviewed cohort required for confirmation. The normal-length causal endpoint therefore remains unestimated.

The certification stream turns the same concern into a finite optimization problem. Given a frozen token tape, original boundaries, admissible spans, a fixed number of units, and a relocation radius k, we minimize the detector score over every legal segmentation in that set. Independent-span scores admit a standard segmentation dynamic program. SemStamp, k-SemStamp, and SimMark instead use first-order relations between adjacent units, requiring an edge-state dynamic program that retains the preceding span. Unlike AliMark's search for the strongest detection evidence among reconstructed candidates, this objective finds the weakest detector score over a declared complete set. The difference concerns the optimization target and coverage contract, not the invention of dynamic programming.

We validate the algorithms against independent exhaustive oracles, implement split-conformal calibration with strict ties, reproduce public detector interfaces on frozen assets, and carry the pipeline to its final resource gate. The terminal SemStamp run is informative precisely because it does not produce a certificate: detector health passes, while exact workload capacity and budget do not. This separates "the detector is unhealthy," "the exact algorithm is wrong," and "the validated implementation is too expensive" as distinct failure modes.

We position the study as a bounded system audit and exact-evaluation framework. Its contributions are:

1. A reproducible A–B–A observation on 30 short SWAN texts, interpreted using the elementary unit-count ceiling of the existing statistic, with complete provenance for the unestimated normal-length endpoint.
2. A precise fixed-token, fixed-unit-count relocation contract and exact DP instantiations for independent-span and first-order adjacent-span scores, with a soundness argument and independent oracle checks.
3. An audited SemStamp/SimMark implementation record that separates correctness, calibration, adapter equivalence, detector health, and resource feasibility. It reports a healthy SemStamp baseline and a score-blind readiness Stop, with no real certified rate.

The known sentence-boundary risk, the one-proportion z-test, its elementary upper bound, standard DP optimality, and split-conformal calibration are background ingredients. The proposed value lies in the specific audited evidence, the declared optimization contract, and the implementation's verified scope; none alone establishes a generally superior watermark.

2 Related Work

Sentence-level semantic watermarks. SemStamp partitions a sentence-embedding space with locality-sensitive hashing and conditions each accepted region on the preceding sentence (Hou et al., 2024a). k-SemStamp replaces that partition with clustering (Hou et al., 2024b). SimMark embeds a signal through

adjacent-sentence similarities and uses a soft detector (Dabiriaghdam & Wang, 2025). SWAN injects AMR templates and detects them through sentence segmentation, neural AMR parsing, graph normalization, S2Match, and document aggregation (Ye et al., 2026).

What the original papers already state. SWAN's Section 3.3 specifies its one-proportion z-statistic, and its Section 6 explicitly names multi-sentence splitting/merging attacks and AMR subgraphs as future directions (Ye et al., 2026). SemStamp's Appendix B defines the corresponding sentence-count z-test; its limitations acknowledge inter-sentence attacks, including reordering (Hou et al., 2024a). SimMark's Section 4.3 already evaluates a Paraphrase+Merge attack (Dabiriaghdam & Wang, 2025). These statements preclude interpreting our short-text observation as a previously unanticipated class of vulnerability.

Resegmentation and rewriting. AliMark's Section 2 analyzes the loss of prefix-conditioned signals after structural perturbations. Sections 3.2.1–3.2.3 specify single-step restructuring candidates, a block-edit alignment DP, and the maximum normalized watermark score over the candidates (Li et al., 2026). SwordStamp is the closest structural neighbor: it evaluates rewording, reordering, and resegmentation, studies unit-count effects, and proposes order-robust sub-sentence units (Diaa et al., 2026). These works preclude claims that we introduce merge/split attacks, boundary exchange, or segmentation-aware detection. In particular, SwordStamp's Section 5.3 explicitly writes the z-score as proportional to $\sqrt{n}$ at a fixed hit rate. Our one-unit ceiling is a direct consequence of that existing statistical form, not a new statistical theorem. Section 5.2 distinguishes AliMark's alignment minimum and outer score maximum from our minimum of the detector score over every declared legal partition. This comparison identifies a problem-specific difference, not a proof of global algorithmic novelty.

DIPPER provides controllable paragraph paraphrasing (Krishna et al., 2023); SIRA targets high-self-information tokens (Cheng et al., 2025); RLCracker studies adaptive worst-case watermark vulnerability (Huang et al., 2026); and Chainwash studies repeated rewriting for diffusion-language-model watermarks (Ameen et al., 2026). WaterPark emphasizes joint reporting of detection, quality, and realized false-positive rates (Liang et al., 2025). FreqMark and CoheMark provide further sentence-level context (Xu et al., 2024; Zhang et al., 2025).

Calibration. We use split-conformal order statistics to freeze thresholds before independent testing (Angelopoulos & Bates, 2023). Binomial uncertainty is reported with one-sided Clopper–Pearson bounds (Clopper & Pearson, 1934). Calibration is part of the certificate procedure: the worst-case score distribution is not silently assigned the raw detector threshold when comparing power at a common target FPR.

3 Study Design and Evidence Separation

3.1 One question, two evidence streams

The unifying question is whether sentence boundaries are an implicit part of a sentence-level watermark's security contract. The empirical stream asks whether changing that interface changes a deployed decision. The certification stream asks whether a detector can guarantee its decision for every segmentation in a declared neighborhood.

The streams appear in one submission because the audit motivates the formal problem and the formalization clarifies what the audit did not establish. They remain separate at the claim level. In particular, no SemStamp or SimMark result upgrades the missing SWAN count-neutral endpoint.

3.2 Fail-closed protocol

Each stage freezes inputs, code identity, thresholds, continuation gates, and resource limits before reading the next class of result. A failed gate terminates that version. We do not delete hard samples, relax a radius, increase a budget, or substitute a cheaper bound while preserving the old version label. This produces many negative engineering records, but prevents partial runs from becoming post-hoc positive claims.

Table 1: Evidence roles in the unified study.

Tier	Evidence	Permitted use
SWAN behavior	30 short-text punctuation A–B–A pairs	System-specific interface result at a historical threshold
SWAN stopping	Two eight-source normal-length pilots	Feasibility and stopping provenance, not detector evidence
Exact algorithms	Oracle and reduction tests on frozen lattices	Correctness for the stated mathematical detector classes
Synthetic statistics	99,980 synthetic null documents	Calibration implementation, not real-system FPR
System reproduction	Frozen public SemStamp/SimMark assets	Implementation equivalence and detector-health diagnosis
Certificate readiness	3,000-row SemStamp workload audit	Capacity and cost conclusion, not a certified robustness rate

4 Part I: Auditing the SWAN Segmentation Interface

4.1 Detection chain and statistical boundary

Let a segmenter map a document to units $s_1, \ldots, s_N$. We distinguish:

- B, SWAN's fixed secret bank of AMR templates;
- $R(s)$, the frozen parser's raw AMR output for unit s;
- $T(R(s))$, the normalized candidate graph supplied to S2Match;
- $D(g_1, g_2)$, an artificial composition of saved graphs used only for scorer diagnostics.

A unit is *green* when at least one fixed-bank match crosses the unit threshold. If G of N units are green, the statistic specified in SWAN's Section 3.3 (Ye et al., 2026) is

$$z = \frac{G - \lambda N}{\sqrt{N\lambda(1-\lambda)}}. \tag{1}$$

Since $0 \le G \le N$, the following elementary bound is immediate:

$$z \le z_{\max}(N) = \sqrt{\frac{N(1-\lambda)}{\lambda}}. \tag{2}$$

Equivalently, a necessary condition for the strict decision $z > \tau$, for $\tau > 0$, is $N > \tau^2\lambda/(1-\lambda)$. This is a corollary of the existing detector formula, not an independent robustness theorem. SwordStamp's explicit $\sqrt{n}$ scaling analysis describes the same unit-count dependence (Diaa et al., 2026).

At $\lambda = 0.25$, $z_{\max} = \sqrt{3} = 1.7321$ for $N = 1$, below the strict threshold $z > 2.33$. SWAN also has two entry-point semantics: the direct-AMR path returns `NaN` for $N < 2$, while the pre-parsed path evaluates Equation 1. We preserve that distinction and never count `NaN` as an attack success.

4.2 Short-text A–B–A intervention

The saved cohort contains 30 independently generated SWAN baselines of 17–46 words and five detection units. The intervention replaces four sentence-final punctuation marks with semicolons without changing lexical tokens or their order. We compare the original text, the punctuation-merged text under default segmentation, and the same merged text supplied with the five saved original spans.

Table 2: Saved SWAN A–B–A behavior at the strict public threshold.

Detector input	Units	$z \leq 2.33$
Original text	5	1/30
Terminal marks replaced by semicolons	1	30/30
Merged text with saved spans restored	5	1/30

Table 3: Normal-length SWAN routes and their terminal evidence.

Route	Structural pass	Detection runs	Terminal interpretation
final1	6/8	0	Auxiliary quality gate failed; no paired cohort
Source-anchored	5/8	0	Structural gate failed before quality review
Target-47, 126 words	1 document	1 baseline	20/20 green, but targeted and not a confirmatory cohort

The paired changes in both directions have exact $p = 3.73 \times 10^{-9}$. Table 2 establishes that external segmentation can reversibly determine the final decision for these texts and this entry point. It does not establish an AMR-specific mechanism: the middle condition changes N from five to one, and the one-unit bound is already sufficient.

A separate no-feedback three-arm study recorded 29/30 valid attacks in its boundary-reorganization arm, but those generated candidates could contain lexical changes. Historical fixed-budget and feedback-assisted studies recorded 191/250 and 218/250 valid attacks, respectively. They motivate the audit but do not replace the controlled A–B–A result.

4.3 Normal-length count-neutral design and stops

The confirmatory intervention would merge one adjacent sentence pair and split another sentence at a detector-blind natural boundary. It preserves tokens, order, punctuation multiset, and unit count: $N(C_2) = N(C_0)$. Quality must be frozen by two independent human reviewers before detection, and each changed detection procedure requires independent null calibration.

The *final1* pilot assigned eight fresh sources. All produced 100–220-word terminal documents, six passed structural checks, and none of those six passed the frozen model-assisted quality labels. It stopped before transformation, parsing, matching, or detection. Those labels are not reported as human quality estimates.

An independent source-anchored route bound ten generation slots to ten source anchors per document. Five of eight documents passed structure, below the six-of-eight continuation gate. All 60 completed slots used fallback, and three additional slots exhausted 30 candidates. It stopped before quality review, transformation, and detection.

A later engineering study showed that one targeted, single-template construction can be both normal length and detectable: a 126-word, 20-unit text had 20/20 strict injections and 20/20 green units ($z = 7.7460$). It lacked source anchoring, independent human review, an independent cohort, and empirical FPR. Its standard count-neutral transformation failed because no admissible natural split was available. An auxiliary equal-N boundary relocation changed 19/19 to 18/19 green, leaving both versions far above threshold. This closes a narrow engineering question, not the confirmatory endpoint.

4.4 SWAN evidence boundary

The normal-length effect beyond unit-count reduction, the contribution of real $R(s)$ and $T(R(s))$, and condition-specific FPR remain unestimated. Artificial graph composition is therefore confined to Appendix A. The finding is a system-interface counterexample, not a general watermark-removal algorithm and not evidence that sentence meaning changed.

5 Part II: Exact Resegmentation Certificates

5.1 Threat set

Let $x = (x_1, \ldots, x_L)$ be a frozen token tape. An original segmentation P_0 has internal boundary set B_0 and fixed unit count $m = |B_0| + 1$. Let $\mathcal{G}$ be the admissible internal gaps, frozen before scores are observed. Each candidate span must satisfy a declared legality predicate, such as a native tokenizer context bound.

For any segmentation P with $|B(P)| = |B_0|$, define the relocation radius

$$\rho(P, P_0) = |B(P) \setminus B_0| = \frac{1}{2}|B(P) \triangle B_0|. \tag{3}$$

The complete radius-k threat set is

$$\mathcal{A}_k(x, P_0) = \left\{ P : \begin{array}{ll} B(P) \subseteq \mathcal{G}, & |B(P)| = |B_0|, \\ \rho(P, P_0) \leq k, & \text{every span in } P \text{ is legal} \end{array} \right\}. \tag{4}$$

The threat model preserves token identity, token order, and unit count. A relocation can move a boundary any distance; it is not a local character-edit metric. The certificate does not cover paraphrasing, insertion, deletion, variable N, arbitrary punctuation rendering, or a natural-language segmenter's reachability.

5.2 Detection, alignment, and worst-case certification

The word "minimum" describes different objects in alignment and certification. We distinguish the original SWAN detector, AliMark's reconstruction/alignment detector, and our finite-set evaluation. Table 4 compares their inputs, objectives, and guarantees. This is a methodological comparison; we do not report matched SWAN–AliMark performance or certify AliMark.

SWAN: scoring the supplied units. For a fixed segmentation P, SWAN marks each unit whose maximum S2Match against the secret bank crosses the unit threshold, then aggregates the hit count with Equation 1 (Ye et al., 2026). Its maximum is over AMR templates for one unit. The original document detector contains neither boundary-search DP nor alignment to a secret sequence. It evaluates the units returned by its segmenter.

AliMark: minimizing block-edit cost to recover evidence. Let $\mathcal{R}(X)$ denote the candidates in AliMark's original Re-Structurer. For an input of N sentences, these are nominally $2N$ candidates: the unchanged text, $N - 1$ single adjacent merges, and N single midpoint splits. They are deliberately limited to one operation (Li et al., 2026, Section 3.2.1). A candidate P has $n(P)$ extracted blocks $b_P^1, \ldots, b_P^{n(P)}$, each of M bits. The Adaptive Bit Sequence Alignment module compares them with ℓ blocks of the secret sequence, where ℓ varies over a declared candidate set $\mathcal{L}(P)$. The reconstructions can change unit count; $\mathcal{R}(X)$ is not our fixed-N relocation set $\mathcal{A}_k$.

For each pair (P, ℓ), AliMark's Equation (4) uses the block-edit DP

$$H_{P,\ell}(u, v) = \min \left\{ \begin{array}{l} H_{P,\ell}(u-1, v) + M, \\ H_{P,\ell}(u, v-1) + M, \\ H_{P,\ell}(u-1, v-1) + \mathrm{Ham}(b_P^u, s^v) \end{array} \right\}. \tag{5}$$

The boundary costs are $H(u, 0) = uM$ and $H(0, v) = vM$. Its block edit rate is

$$r(P, \ell) = \frac{H_{P,\ell}(n(P), \ell)}{M \max\{n(P), \ell\}}. \tag{6}$$

Using the original paper's length-indexed null mean and standard deviation, the final detector is

$$\mathcal{Z}_{\mathrm{Ali}}(X) = \max_{\substack{P \in \mathcal{R}(X) \\ \ell \in \mathcal{L}(P)}} \frac{\bar{\mu}_{M,n(P)} - r(P, \ell)}{\bar{\sigma}_{M,n(P)}}. \tag{7}$$

Table 4: Different optimization targets. Exactness is relative to the stated search object, not to arbitrary meaning-preserving edits.

Dimension	Original SWAN	Original AliMark	This exact evaluation
Search object	AMR bank for each supplied unit	Limited text reconstructions, secret-prefix lengths, block alignments	Every legal partition in a fixed-token, fixed-N relocation set
DP state	No document-level DP	Consumed extracted blocks and secret blocks (u, v)	Units, boundary budget, token endpoint; previous span for pair scores
Inner objective	Maximum bank similarity	Minimum block-edit cost	Minimum additive or adjacent-span detector sum
Document output	Hit-count z-score on one segmentation	Maximum normalized score across reconstruction/alignment candidates	Minimum detector score across all declared legal partitions
Meaning of success	Supplied units exceed the threshold	A tested reconstruction/alignment exceeds the threshold	All partitions in the declared threat set exceed the threshold
Scope limit	No partition-uniform guarantee	Alignment optimum is not an all-partition certificate	No paraphrase, variable-N, or AliMark certificate claimed

Equations 5–7 restate AliMark's Equations (4)–(6), with renamed indices (Li et al., 2026). For fixed (P, ℓ), DP finds the least-cost alignment; the outer maximum selects the strongest normalized evidence. Because normalization varies with candidate length, that outer maximum need not equal a single global minimum of unnormalized edit cost. Its decision asks whether a reconstruction and an alignment provide sufficient evidence. An optimal alignment does not imply that every possible text partition was searched or remains detectable.

Our evaluation: minimizing the detector score. Here the original token tape is fixed, each DP path constructs a partition in $\mathcal{A}_k(x, P_0)$, and the output is

$$C_k(x) = \min_{P \in \mathcal{A}_k(x, P_0)} D(x, P). \tag{8}$$

For a nonempty finite threat set and a fixed threshold, the implication $C_k(x) > \tau \Rightarrow \forall P \in \mathcal{A}_k,\ D(x, P) > \tau$ follows from the definition of a minimum. The substantive algorithmic obligation is to show that the implemented DP returns exactly that minimum. It chooses text boundaries and, for adjacent-span scores, retains the previous span. It does not align extracted bit blocks to a secret sequence. AliMark's higher-is-better aggregate in Equation 7 includes candidate selection and alignment; it is not a locally decomposable D covered by the DP classes below without another derivation.

Thus the contribution concerns a specified worst-case evaluation problem and its verified solution. Standard block-edit DP and standard optimal-substructure reasoning are prior ingredients; this comparison does not establish that exact resegmentation is globally novel or practically non-vacuous.

5.3 Detector classes

For independent-span scores $s(i, j)$,

$$D(x, P) = b + \frac{1}{m} \sum_{[i,j) \in P} s(i, j). \tag{9}$$

For first-order adjacent-span detectors, let $0 = b_0 < b_1 < \cdots < b_m = L$ be the segmentation endpoints and let $q(h, i, j)$ score adjacent spans $[h, i)$ and $[i, j)$. Then

$$D(x, P) = b + \frac{1}{c} \sum_{t=2}^{m} q(b_{t-2}, b_{t-1}, b_t), \qquad c > 0, \tag{10}$$

where b, c, model parameters, and threshold are fixed over the threat set. SemStamp and k-SemStamp use adjacent region transitions; SimMark uses a soft count derived from adjacent sentence-embedding similarity. For fixed m, their offsets and positive normalizers fit Equation 10.

The formulation excludes global attention, higher-order unit interactions, scan/post-selection statistics, stochastic scores, and thresholds that adapt to the candidate segmentation.

5.4 Exact dynamic programs

For independent spans, let $F(t, j, r)$ be the minimum score sum for segmenting the first j tokens into t units while introducing r non-original internal boundaries. The recurrence adds a legal span $[i, j)$ and increments r when the internal endpoint j is not in B_0. With maximum span length W, the time complexity is $O(m(k+1)LW)$ after span scores are available.

Pairwise detectors require the previous span. Let $F(t, h, i, r)$ be the minimum pair-score sum for the first i tokens split into t units, with final span $[h, i)$ and r introduced boundaries. Adding $[i, j)$ gives

$$F(t+1, i, j, r + \delta(j)) = \min_h \left[F(t, h, i, r) + q(h, i, j) \right], \tag{11}$$

where $\delta(j) = \mathbf{1}[j \notin B_0]$ for $j < L$ and $\delta(L) = 0$. Only legal gaps, legal spans, $t \le m$, and $r \le k$ are retained. The exact worst score is

$$C_k(x) = b + \frac{1}{c} \min_{\substack{h \\ 0 \le r \le k}} F(m, h, L, r). \tag{12}$$

The pairwise time and predecessor-space bounds are $O(m(k+1)LW^2)$ and $O(m(k+1)LW)$, respectively; $k+1$ includes the zero-edit state.

Soundness. This is the standard optimal-substructure argument specialized to the declared state and legal-path constraints. Every path accepted by Equation 11 contains exactly m legal spans, and its count r equals the number of introduced boundaries. Conversely, every $P \in \mathcal{A}_k$ induces exactly one such path. The recurrence keeps the minimum score among paths with the same sufficient state. By induction on t, Equation 12 is therefore the minimum over $\mathcal{A}_k$. Thus

$$C_k(x) > \tau \quad \Longrightarrow \quad \forall P \in \mathcal{A}_k(x, P_0),\ D(x, P) > \tau. \tag{13}$$

This guarantee is exact for the declared lattice and detector class; it says nothing about partitions outside that contract.

5.5 Calibration

Because $\mathcal{A}_k$ contains P_0 and expands with k,

$$C_{k+1}(x) \le C_k(x) \le C_0(x). \tag{14}$$

Using a fixed raw upper-tail threshold cannot increase null FPR, but can make the certificate unnecessarily conservative. For a power comparison at a common target FPR α, we calibrate each complete C_k on an independent null split. For n calibration scores, the strict threshold is the r-th ascending order statistic,

Table 5: Model-independent validation of the exact framework.

Check	Scale	Result
Independent-span DP	7,691 cases	Exact agreement with brute force
Pairwise DP	7,669 cases	Exact score, count, decision, and witness
Pairwise-to-additive reduction	1,000 cases	Exact agreement
300-token pairwise case	14,606,509 partitions	39,977 states, 449,337 transitions, 0.8054 s
Synthetic calibration	99,980 nulls	All 80 family/replicate/radius FPR gates passed

$$r = \lceil (n+1)(1-\alpha) \rceil, \qquad \tau_k = C_{k,(r)}. \tag{15}$$

Ties remain on the non-detection side because the decision is $C_k > \tau_k$. A separate test null estimates realized FPR with a one-sided Clopper–Pearson upper bound. Positive certification is reported with a one-sided lower bound. These quantities are not pooled across radii.

6 Algorithm and Systems Validation

6.1 Oracle and synthetic calibration results

The pairwise total in Table 5 comprises 5,669 systematic and 2,000 random tied-score instances. The independent-span total comprises 5,691 systematic and 2,000 random instances and checks 36,824 legal partitions. All witness scores are independently recomputed; $k = 0$ returns only the original segmentation; illegal or missing inputs fail closed.

In the synthetic calibration study, separately calibrated continuous-null FPR means ranged from 4.57% to 5.18% over $k = 0, \ldots, 3$. The largest one-sided 95% FPR upper bound among 80 checks was 7.465%, below the frozen 8% gate. Reusing the raw τ_0 at $k \geq 1$ yielded zero FPR in these synthetic families, illustrating conservatism rather than a real-watermark result.

6.2 Adapters and compiled core

The real-system adapter preserves a typed unit-list tape and reconstructs the original units byte-for-byte at $k = 0$. Public-code inspection showed that SemStamp, k-SemStamp, and SimMark are first-order adjacent-span systems, not independent-span systems. Frozen small-sample tests then matched official and direct computations: SemStamp passed 258/258 unit, region, mask, hit, and score checks on 43 eligible documents; SimMark passed 240/240 preprocessing, cosine, soft-count, and score checks on 60 documents. The SimMark check used an explicitly diagnostic $\gamma = 0.5$ sentinel and is not a detector-health result.

The Python sparse implementation exceeded a 60-second gate on the largest frozen real-data shape. A C++17 grouped, layer-local core preserved all oracle outputs and processed a shape with 14,169 atoms, 845 units, and $k = 3$ in 18.35 and 18.47 seconds using about 165 MB. It represented 39,353,742,686,409,367,909 legal segmentations through 961,430 unique spans and 38,664,070 unique pair queries.

Materializing pair tables then failed a five-million-pair gate. A factorized core instead caches one feature per span and computes pair values on demand. It passed real-provider equivalence for 12 documents per system, but full two-system score-blind projections were 22.44 GPU-hours for SemStamp and 71.81 GPU-hours for SimMark. Increasing SimMark batch size gave only 1.0027× speedup, and an exploratory global-reuse audit reduced work by only 0.6748%; that audit also violated its frozen row-scope contract and is marked invalid.

6.3 Detector reproduction and health

SimMark v9 completed exact scores for 200 fit and 1,800 calibration/test/positive documents over $k = 0, \ldots, 3$, but the official histogram produced $\gamma = 0$ and clean-positive detection was only 8.9%. Subsequent

Table 6: Real-system terminal results. Certificate rates are intentionally absent when detector health or readiness failed.

Run	Test FPR	Positive rate	Interpretation
SimMark v9, $k = 0$	27/400	89/1000	Published-asset baseline unhealthy; no robustness inference
SimMark v9, $k = 1$	37/400	97/1000	Recalibrated exact score, but invalid as watermark evidence
SemStamp v10, $k = 0$	33/400	974/999	TPR passed; FPR CP95 upper 10.88% failed the 8% gate
SemStamp v17, $k = 0$	47/1000	954/1000	Detector-health Go; CP95 bounds 5.95% and 94.16%
SemStamp v18, $k \geq 1$	–	–	Readiness Stop before score or certificate computation

Table 7: Frozen SemStamp v18 readiness gates.

Gate	Limit	Observed maximum	Result
Pruned pairs per document	50,000,000	78,642,945	Stop
All-allowed pairs per document	50,000,000	83,540,900	Stop
Two-replay budget	72 h	86.6501 h	Stop
Rows reconstructed	3,000	3,000	Pass

diagnosis reproduced all 2,000 official/direct document scores and localized the discrepancy to Instructor prompt-pooling compatibility. A no-prompt-pooling diagnostic restored 99.0% positive detection on the frozen split, but it was not preregistered as a new detector and did not retroactively validate v9.

SemStamp v10 detected 974/999 positives, but its test FPR was 8.25% and the one-sided 95% upper bound was 10.88%, above the frozen 8% gate. Across 100 deterministic rotated splits, this split was near the high end of the observed FPR distribution. Rather than choose a favorable split, we constructed fresh, identity-separated C4 cohorts and a new positive set.

The positive-generation sequence itself exposed reproducibility failures: an initial smoke produced two prompt-only samples; one runner stalled because its fallback was unreachable; a bounded replacement broke a strict-prefix contract; and an independent audit later found five multi-sentence prompt-only outputs among 1,000 claimed positives. A disclosed post-hoc validation rescue retained 995 valid outputs and generated five replacements from the fixed queue. This history is not rewritten as preregistered success.

With the resulting 1,000 positives and two independent 1,000-document null cohorts, SemStamp v17 passed detector health. The threshold was the 951st calibration order statistic, $\tau_0 = 1.6667$. Test FPR was 47/1000 (CP95 upper 5.95%), positive TPR was 954/1000 (CP95 lower 94.16%), and two 3,000-row replays were byte-identical. Official and direct scores agreed exactly on 6,000 evaluations.

6.4 Final exact-certificate readiness

SemStamp v18 froze the same 3,000 identities, native tokenizer legality, $k = 0, \ldots, 3$, a 50-million-pair per-document cap, and a 72 GPU-hour budget for two complete replays. A score-blind pass reconstructed every workload without loading the detector.

The automatic gate left formal B1 as `STOPPED / Not started`. Detector, model, region, score, threshold, generation, and certificate calls were all zero. No threshold τ_k, witness, test FPR, positive certified rate, or non-vacuity statistic exists for v18. The result is a capacity and budget limit of the validated exact implementation, not evidence that SemStamp is or is not robust to resegmentation.

7 Discussion

7.1 Contribution beyond the known risk

SWAN already anticipates multi-sentence attacks, and AliMark and SwordStamp already analyze and mitigate structural perturbations. The paper's focus is therefore the evidence added by a reproducible case study, an explicit optimization contract, and a system audit. The A–B–A result measures a specific reversible behavior; its statistical explanation is elementary. The DP supplies a complete minimum for stated local detector classes; its proof uses standard principles. The resource study distinguishes a healthy detector from an executable certificate on the frozen workload.

These contributions remain limited. The available behavior does not establish an AMR-specific mechanism at normal length, and the exact framework has no final real certified rate. The work should be assessed as a bounded audit and exact-evaluation study, rather than as a new watermark, a previously unknown attack class, or a general impossibility theorem. The number of engineering iterations is not itself evidence of novelty.

7.2 What the unified evidence establishes

The SWAN observation shows that segmentation is not merely preprocessing: it can determine the final decision. The exact framework then states what a stronger robustness claim would require: enumerate a declared finite boundary lattice implicitly, minimize the complete detector statistic, and calibrate that statistic on independent null data.

The algorithmic result is positive. For independent-span and first-order pairwise detectors, exhaustive search over all legal partitions can be replaced by exact dynamic programming without weakening the threat set. The systems result is mixed. Detector semantics, data identity, rendering, feature extraction, calibration, and resource limits all matter before a certified rate can be interpreted. On our final SemStamp assets, detector health was adequate but the validated exact implementation exceeded its predeclared capacity and runtime gates.

7.3 Why the two streams do not substitute for each other

The certificate threat set fixes token identity, token order, and unit count. The strongest SWAN behavior result changes unit count. Conversely, the SemStamp and SimMark detectors do not use SWAN's AMR parser or fixed AMR bank. The common problem is the segmentation interface. The contribution is the specific evidence and evaluation contract; causal and statistical claims remain system-specific.

7.4 Implications for watermark evaluation

Sentence-level watermark reports should specify at least:

1. the segmentation and rendering contract;
2. whether unit count may change under the threat model;
3. whether the detector statistic decomposes locally;
4. whether thresholds are raw or calibrated after robustification;
5. clean TPR and independent realized FPR before robust conclusions;
6. the coverage and computational cost of any claimed certificate.

Reporting only a successful attack or only a dynamic program is insufficient. The former may collapse statistical power; the latter may be exact but computationally or statistically vacuous.

8 Limitations

- **Originality scope.** Sentence-boundary risks, unit-count scaling, reconstruction/alignment, and standard DP and calibration principles have prior sources. We do not establish global algorithmic novelty, and the bounded case study does not resolve the missing mechanism or real-certificate endpoints.
- **AliMark comparison.** Section 5.2 compares objectives and search coverage. It is not a performance benchmark, a criticism of AliMark's reported FPR, or a robustness certificate for its reconstruction/alignment detector.
- **SWAN sample scope.** The A–B–A cohort contains 30 English texts of 17–46 words and five original units. The normal-length, count-neutral endpoint was not run.
- **SWAN quality and calibration.** The short-text result lacks condition-specific FPR. The normal-length routes did not produce the independently human-reviewed cohort required by protocol.
- **Threat-set scope.** The certificate fixes tokens, order, and unit count. It does not certify arbitrary paraphrase, punctuation realizability, variable N, or semantic equivalence.
- **Detector class.** Exactness applies to independent-span or first-order adjacent-span scores with fixed affine aggregation. It does not cover global or higher-order detectors.
- **Real certificate endpoint.** SimMark's completed run had an unhealthy baseline. SemStamp's final healthy run stopped at readiness. No defensible real certified rate is reported.
- **Assets and environments.** Results depend on frozen public revisions, CUDA behavior, tokenizers, and generated SemStamp positives. The v16 positive set is a disclosed post-hoc validation rescue.
- **Multiplicity.** Radius-specific intervals are not presented as a simultaneous confidence statement over all radii.

Broader Impact Statement

This work is intended to improve the evaluation and engineering of text watermarks by making segmentation assumptions explicit. The same analysis could inform attempts to reduce detectability. We limit that risk by reporting aggregate frozen results, a bounded threat model, and no live key, credential, sealed text, or service-targeting procedure. Exact certificates, when available, can support defenders as well as auditors.

9 Conclusion

The known dependence on sentence boundaries is part of the operational contract of sentence-level watermarks. SWAN's decision changed reversibly under a short-text segmentation intervention, but the available evidence does not isolate an AMR-specific or normal-length count-neutral mechanism. Exact dynamic programming can certify all segmentations in a fixed-token, fixed-unit-count relocation ball for local detector classes, and our implementation passes exhaustive correctness and calibration checks. The final real-system result is nevertheless a Stop: SemStamp detector health passed, while exact-certificate capacity and budget did not. The resulting contribution is a bounded behavioral audit, a verified exact evaluation for a declared threat set, and an implementation feasibility record. It does not introduce the underlying boundary risk, the z-test, or standard DP principles.

References

Mohd Ruhul Ameen, Akif Islam, Nadim Mahmud, and Md. Ekramul Hamid. Chainwash: Multi-step rewriting attacks on diffusion language model watermarks. *arXiv preprint arXiv:2605.05503*, 2026. URL `https://arxiv.org/abs/2605.05503v1`.

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101.

Yixin Cheng, Hongcheng Guo, Yangming Li, and Leonid Sigal. Revealing weaknesses in text watermarking through self-information rewrite attacks. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. URL https://openreview.net/forum?id=fE3kgW7kMp.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

Amirhossein Dabiriaghdam and Lele Wang. SimMark: A robust sentence-level similarity-based watermarking algorithm for large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. URL https://arxiv.org/abs/2502.02787v2.

Abdulrahman Diaa, Jonathan Petit, and Florian Kerschbaum. Semantic watermarking with order-robust detection over sub-sentence units. *arXiv preprint arXiv:2608.27666*, 2026. URL https://arxiv.org/abs/2608.27666v1.

Abe Hou, Jingyu Zhang, Tianxing He, Yichen Wang, Yung-Sung Chuang, Hongwei Wang, Lingfeng Shen, Benjamin Van Durme, Daniel Khashabi, and Yulia Tsvetkov. SemStamp: A semantic watermark with paraphrastic robustness for text generation. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 1706–1715, 2024a. doi: 10.18653/v1/2024.naacl-long.226.

Abe Bohan Hou, Jingyu Zhang, Yichen Wang, Daniel Khashabi, and Tianxing He. k-SemStamp: A clustering-based semantic watermark for detection of machine-generated text. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024b. URL https://arxiv.org/abs/2402.11399v2.

Hanbo Huang, Yiran Zhang, Hao Zheng, Xuan Gong, Yihan Li, Lin Liu, Zhuotao Liu, and Shiyu Liang. RLCracker: Evaluating the worst-case vulnerability of LLM watermarks with adaptive RL attacks. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2509.20924v2.

Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2303.13408v2.

Yuexin Li, Wenjie Qu, Linyu Wu, Yulin Chen, Yufei He, Tri Cao, Bryan Hooi, and Jiaheng Zhang. AliMark: Enhancing robustness of sentence-level watermarking against text paraphrasing. *arXiv preprint arXiv:2605.29434*, 2026. URL https://arxiv.org/abs/2605.29434v1. Accepted at ICML 2026.

Jiacheng Liang, Zian Wang, Lauren Hong, Shouling Ji, and Ting Wang. Watermark under fire: A robustness evaluation of LLM watermarking. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. URL https://arxiv.org/abs/2411.13425v4.

Zhenyu Xu, Kun Zhang, and Victor S. Sheng. FreqMark: Frequency-based watermark for sentence-level detection of LLM-generated text. *arXiv preprint arXiv:2410.10876*, 2024. URL https://arxiv.org/abs/2410.10876v1.

Ziping Ye, Gourab Dey, Christos Christodoulopoulos, Charith Peris, Anil Ramakrishna, Weitong Ruan, Aram Galstyan, Kai-Wei Chang, Rahul Gupta, and Ninareh Mehrabi. SWAN: Semantic watermarking with abstract meaning representation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 36304–36315, 2026. doi: 10.18653/v1/2026.acl-long.1681.

Junyan Zhang, Shuliang Liu, Aiwei Liu, Yubo Gao, Jungang Li, Xiaojie Gu, and Xuming Hu. CoheMark: A novel sentence-level watermark for enhanced text quality. *arXiv preprint arXiv:2504.17309*, 2025. URL https://arxiv.org/abs/2504.17309v1.

A SWAN Exploration and Diagnostic Results

Exploration history. Plans A–E explored ordinary paraphrasing, AMR node and edge edits, modifier and argument operations, reentrancy changes, sentence restructuring, and feedback-assisted candidate selection. Increasingly strong graph edits traded off detectability against factual preservation. Plan F corrected denominators to quality-gated valid attacks. Plans G–J isolated sentence boundaries, tested no-feedback selection, compared SIRA-adapted, and added unmatched KGW and SynthID-Text controls. These studies motivate the interface audit but are not pooled as one confirmatory endpoint.

Artificial-graph diagnostic. Across 108 accepted injection targets, median target S2Match F1 was 0.8197 for original candidates, 0.4502 for artificial compositions, and 0.5200 for saved merged-text reparses. Target-green counts were 108/108, 0/108, and 8/108. Artificial composition adds a root and connector edges and is not a natural-text counterfactual.

Stopped defenses and range controls. Unmatched boundary-reorganization controls recorded valid attack rates of 0/30 for KGW and 1/30 for SynthID-Text. Three local SWAN defenses failed their development gates: multi-instance rules recovered 0/15 across three representations; template-tail detection reached 4/15 on original graphs and 2/15 on reparses, below a joint 8/15 gate; relation-motif recall at five was 41/56 and 33/56, each one below its continuation gate. A direct resegmentation mitigation changed valid escapes from 23/30 to 19/30 ($p = .125$) while human-null false positives changed from 3/250 to 5/250. No effective segmentation-invariant SWAN defense is claimed.

B Certificate Engineering Timeline

Table 8: Terminal certificate stages. "Pass" applies only to the stated gate, not to an end-to-end robustness claim.

Stage	Purpose	Terminal evidence
WP1	Additive exact DP	7,691 oracle cases passed
WP1b	Pairwise edge-state DP	7,669 oracle cases and 1,000 reductions passed
WP1c	Certificate calibration	99,980 synthetic nulls; 80/80 FPR gates passed
WP2a	Canonical adapter	500 model-free instances and $k = 0$ restoration passed
v1–v2	Public assets and lossless adapter	Official/direct checks passed; frozen eligibility contracts failed
v3	Typed structural separator	Eligibility recovered; Python core timed out
v4	Compiled sparse core	Algorithm-Ready on the largest frozen shape
v5	External pair table	Stopped at 5,000,001 pairs before model loading
v6	Factorized features	Provider equivalence passed; SimMark projection 71.81 h
v7	Batch scaling	Best speedup 1.0027×; Stop
v8	Global reuse	Invalid row scope; exploratory reuse only 0.6748%
v9	SimMark long run	Exact scores completed; detector baseline unhealthy
v10	SemStamp health-first	TPR passed; independent FPR bound failed
v11	Reproduction diagnosis	Official/direct equivalence; SimMark pooling and SemStamp split instability
v12	Fresh identities	12,612 legal identities; smoke 18/20, then Stop
v13	Replacement-aware generation	Stalled at attempt 372 after 84,981 s
v14	Bounded fallback	Strict-prefix divergence at attempt 2; Stop
v15	Causal-prefix generation	1,000 claimed; audit found five prompt-only rows
v16	Validation rescue	Disclosed 995+5 cohort; identity audit Go
v17	SemStamp detector health	4.7% FPR, 95.4% TPR; Go
v18	Exact-certificate readiness	78.6M pairs and 86.65 h; Stop before scoring

No stopped version is retroactively relabeled as successful. The v9 scores remain a detector-reproduction diagnostic; v10 remains a failed health gate; v16 remains post-hoc; and v18 contains no certificate output.

C Additional SimMark Diagnostics

The v9 radius-specific diagnostic rates were:

k	Independent-test FPR	Positive rate
0	27/400	89/1000
1	37/400	97/1000
2	36/400	64/1000
3	36/400	34/1000

Each radius used its own calibrated threshold, so the positive point estimate need not be monotone even though every document satisfies $C_0 \geq C_1 \geq C_2 \geq C_3$. The table is not a certified-robustness result because the $k = 0$ detector failed before-attack health.

D Unestimated Endpoints

Table 9: Important quantities that must not be read as zero.

Endpoint	Status
SWAN normal-length N-preserving effect	Not run
SWAN parser and candidate-graph mechanism	Not run
SWAN condition-specific FPR	Not run
SemStamp exact C_k on final v18 cohorts	Not run
SemStamp post-certificate τ_k, FPR, TPR	Not run
SemStamp real certified-positive rate	Not run
Healthy SimMark certified-positive rate	Not available
Cross-system certified robustness comparison	Not available

E Reproducibility Boundary

The internal evidence tree stores frozen protocols, code and model revisions, identity manifests, row-level derived outputs, independent audits, terminal states, and SHA256 chains. The SWAN reconciliation audit preserves 39,457 file hashes and 82 original-document copies. The SemStamp v17 health run uses two byte-identical 3,000-row replays; the v18 workload and timing rows are independently recomputed without loading the detector.

A public artifact must exclude credentials, sealed source or generated text, and third-party payloads without redistribution permission. It can include the claim registry, aggregate tables, exact-DP and audit code, frozen revisions, non-sensitive manifests, and synthetic tests. Local paths are reproducibility indices, not publication URLs.